

Applications and limitations of pluripotent stem cell-derived liver organoids

Zhiwei Feng¹, Bingrui Zhou¹, Qizhi Shuai¹, Jianbing Mu², Jun Xie¹

¹Shanxi Key Laboratory of Birth Defect and Cell Regeneration, Department of Biochemistry and Molecular Biology, MOE Key Laboratory of Coal Environmental Pathogenicity and Prevention, Shanxi Medical University, Taiyuan, Shanxi 030001, China;

²Laboratory of Malaria and Vector Research, National Institute of Allergy and Infectious Diseases, National Institutes of Health, Rockville, MD 20852, United States.

Abstract

Liver disease is one of leading causes of death worldwide. However, current liver models have limited research progress. Therefore, models that accurately reflect the physiological functions of the human liver are urgently needed in both clinical and laboratory research. Over the past decade, liver organoids have emerged as valuable research tools offering significant breakthroughs and demonstrating great promise as advanced liver models. Liver organoids derived from pluripotent stem cells (PSCs), including embryonic and induced pluripotent stem cells, have shown significant potential for modeling liver diseases and drug responses. These miniature three-dimensional (3D) structures replicate the complexity of the liver and offer a platform for studying liver development and disease progression. The ability to create personalized organoids from patient-derived cells paves the way for precision medicine and drug screening. Owing to the pluripotency of PSCs, PSC-derived liver organoids (PSC-LOs) closely mimic the true structure of the liver and offer a wide range of applications. With advances in research, bioengineered liver organoids have the potential to revolutionize regenerative medicine, disease modeling, and the understanding of liver pathophysiology. This review provides an overview of liver organoid development and discusses their applications in liver regeneration, disease modeling, drug screening, toxicity assessment, organ transplantation, and regenerative medicine. Finally, we discuss the limitations and future development directions of PSC-LOs.

Keywords: Liver organoids; Pluripotent stem cells; Generation; Application; Limitation

Introduction

The liver is the largest metabolic organ in the human body, playing an important role in various physiological processes and maintaining homeostasis. Liver has a dual blood supply system: the hepatic artery provides oxygen and the portal vein carries substances from the gastrointestinal tract for metabolic processing. In the liver, nutrients are metabolized into energy, while non-nutritional substances such as drugs, toxins, and some metabolic products are inactivated, decomposed, or converted for excretion. Due to its exposure to hepatotoxic substances, the liver is prone to damage, leading to diseases such as drug-induced hepatitis, liver fibrosis, cirrhosis, and biliary tract diseases. At present, liver disease poses a heavy health burden worldwide.^[1] Research on liver diseases has always been constrained by existing liver models. The current ideal model for *in vitro* evaluation of drug metabolism and liver toxicity is two-dimensional (2D) cultured primary human hepatocytes (PHHs).^[2] However, PHHs are difficult to procure, exhibit a loss of proliferation capacity *in vitro*,

and progressively fail to sustain cellular structure and function over extended periods.^[2] Additionally, cell lines like HL-7702, WRL-68, and CCL13, previously considered normal human liver cell lines, which be contaminated with HeLa cells, complicating their use.^[3] These 2D cultures cannot replicate the liver's multicellular complexity and 3D structure, limiting their ability to predict drug reactions and disease mechanisms accurately. Animal models, while commonly used, have significant genetic and metabolic differences from humans, making clinical translation of findings challenging.^[4]

Recent advancements in organoid culture systems have addressed some limitations of 2D cultures and animal models. Compared with 2D culture models, organoids can simulate the 3D structure of the original tissue and preserve the genetic characteristics and heterogeneity, offering a promising model system.^[5,6] Multiple laboratories have established various liver organoids through different methods, which are widely used in liver disease research, yielding positive results.

Access this article online

Quick Response Code:



Website:
www.cmj.org

DOI:
10.1097/CM9.0000000000003707

Correspondence to: Jun Xie, M.D., Ph.D. Shanxi Key Laboratory of Birth Defect and Cell Regeneration, Department of Biochemistry and Molecular Biology, Shanxi Medical University, Taiyuan, Shanxi 030001, China
E-Mail: junxie@sxmu.edu.cn

Copyright © 2025 The Chinese Medical Association, produced by Wolters Kluwer, Inc. under the CC-BY-NC-ND license. This is an open access article distributed under the terms of the Creative Commons Attribution-Non Commercial-No Derivatives License 4.0 (CCBY-NC-ND), where it is permissible to download and share the work provided it is properly cited. The work cannot be changed in any way or used commercially without permission from the journal.

Chinese Medical Journal 2026;139(3)

Received: 26-07-2024; Online: 28-07-2025 Edited by: Xiangxiang Pan

Overview of Liver Organoids

The concept of organoids has been in existence for decades and is originally defined as various 3D culture systems that resemble organs to varying extents.^[6] A significant milestone was achieved in 2009, when Hans Clevers' lab established a mouse gut organoid system. They cultured Lgr5-positive adult mouse intestinal stem cells in Matrigel and revealed that these cells self-organize and differentiate into intestinal crypt-villus structures without a mesenchymal niche.^[7] This study marked the beginning of the modern era of organoid research.^[7] Subsequently, organoids of human organs and tissues, such as the liver,^[8] brain,^[9] stomach,^[10] kidney,^[11] retina,^[12] and lungs,^[13] were successively established and reported. Organoids are defined as *in vitro* 3D cell clusters derived from tissue-resident stem/progenitor cells (including tumor stem cells), embryonic stem cells (ESCs), or induced pluripotent stem cells (iPSCs) that are capable of self-renewal and self-organization, mimicking some structures and functions of the originating tissue.^[14]

The first liver organoids were reported in 2013 by Hans Clevers and Takanori Takebe.^[8,15] Huch *et al*^[15] cloned and expanded a single Lgr5⁺ liver duct cell from adult mice into a cystic bile duct organoid expressing Krt19 and Krt7 over a long period of time. The cells in these organoids expressed multiple hepatocyte lineage and bile duct markers. Under certain stimulatory conditions, they express markers of mature liver cells and perform certain functions in the liver. Subsequently, they cultured EpCAM⁺ bile duct cells from human liver biopsy tissues to form organoids that could be amplified and passaged. After multiple passages, the organoids still maintained genetic stability, expressed liver-specific markers, and had functions such as albumin secretion, low density lipoprotein (LDL) uptake, and glycogen storage.^[16] Takebe *et al*^[17] co-cultured human iPSC-derived liver endodermal cells with human umbilical vein endothelial cells (HUVECs) and mesenchymal stem cells (MSCs), creating liver organoids or liver buds that integrated with the host vasculature and secreted human albumin when transplanted into mice. These liver bud tissues can be transplanted into non-obese diabetic/severe combined immune deficiency (NOD/SCID) mouse models, where they establish a vascularized network connected to the host's blood vessels. Additionally, they secrete human albumin into the mouse circulatory system. This method was later improved to differentiate hepatic endoderm cells, HUVECs, and MSCs from hiPSCs for liver bud establishment. Using an Omni-well-array culture platform, homogeneous and miniaturized liver buds can now be produced on a clinical scale. Consequently, multiple laboratories worldwide have reported various protocols for establishing liver organoids. There have been an increasing number of reports on liver organoids. Supplementary Figure 1, <http://links.lww.com/CM9/C518> shows the number of publications with the terms "liver organoids" in their title, since 2013 and up until 2024, based on searches in PubMed.

Liver organoids can be classified into two types based on their source.^[18] One type was derived from stem/

progenitor cells (including tumor stem cells) obtained from liver biopsy or liver resection tissues that is similar to the methods used in the Hans Clevers' laboratory. The organoids produced by this method have the advantages of a short culture cycle, high maturity, good genome stability, and high differentiation efficiency.^[18] However, due to the limited stemness of the source cells, the cellular components in these liver organoids are relatively simple, including only epithelial cell types such as hepatocytes and/or cholangiocytes, or one type of tumor cells.^[18] The other organoids production method, similar to Takebe's method, uses PSCs, including ESCs and iPSCs, which progress through stages of endodermal induction, liver specialization, and cell maturation to generate liver organoids, pluripotent stem cell derived liver organoids (PSC-LOs).^[18] These PSC-LOs can contain various liver cellular components, closely resembling the true liver structure.^[18] Thus, most liver organoids reported in the literature are derived from the PSC differentiation.^[17,19-26]

Generation of PSC-LOs

The liver originates from the endoderm and five key pathways play important roles in its development, retinoic acid (RA), wingless/integrated (WNT), transforming growth factor- β (TGF- β), protein kinase A system (PKA), and bone morphogenetic proteins (BMP) pathways.^[27] These pathways function at different stages of liver development.^[27] By adding relevant cytokines and active compounds to the culture medium, PSCs can be induced to differentiate into liver progenitor cells through the definitive endoderm, posterior foregut, and hepatoblasts stages [Figure 1].^[28]

Takebe *et al*^[17] pioneered the use of PSCs to generate 3D liver organoids by co-culturing PSCs-derived liver progenitor cells with HUVECs and MSCs to form liver organoids, known as liver buds (LBs). These LBs contain various liver cellular components and can generate blood vessels connected to the host's circulatory system after transplantation into immunodeficient mice. They later refined the system so that all cells in the 3D culture environment originated from PSCs. In 2019, Pettinato and colleagues co-cultured human pluripotent stem cells (hPSCs) with human adipose microvascular endothelial cells (HAMECs) to establish a similar liver organoids system.^[29] Another study in the same year involved 3D co-culturing human embryonic stem cells (hESCs)-derived liver stem/progenitor cells with human fetal liver mesenchymal cells (hFLMCs) to form liver organoids with hepatic stromal components.^[30]

Some studies have focused on differentiating PSCs into live organoids containing multiple cellular components. For example, Guan *et al*^[31] differentiated hPSCs into liver progenitor cells and subsequently produced liver organoids composed of hepatocyte-like cells (HLCs) and bile duct cell-like cells (CLCs) by adding exogenous growth factors to the Matrigel. The liver organoids can self-renew and perform liver functions such as glycogen storage, liver-specific drug metabolism, and albumin and bile secretion.^[31] In another study, Wu *et al*^[32] induced differentiation of the endoderm and mesoderm by adding

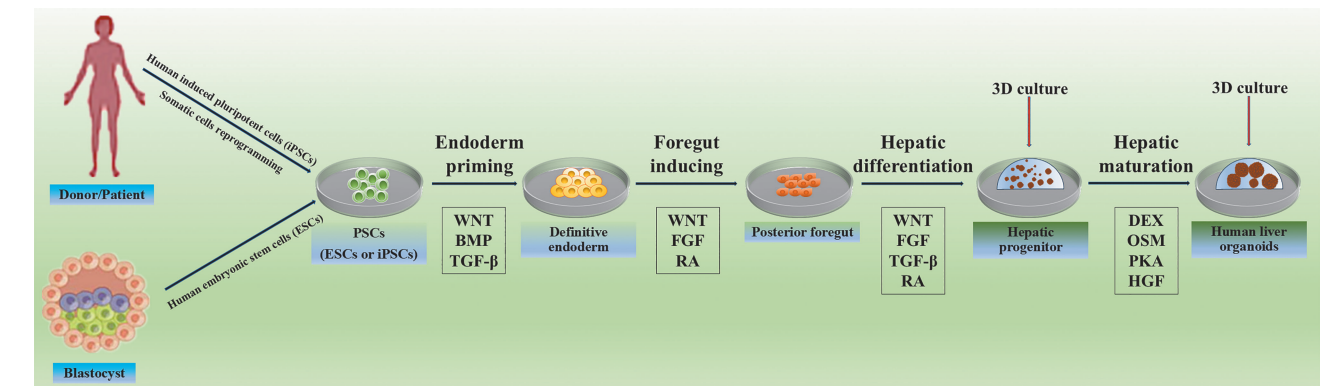


Figure 1: Generation of PSC-LOs. Currently, many different protocols are available to generate PSC-LOs. Overall, by adding relevant cytokines and active compounds to the culture medium to simulate the process of liver development *in vivo*, PSCs can be induced to differentiate into liver progenitor cells *in vitro* through the definitive endoderm, posterior foregut, and hepatoblasts stages. Subsequently, in the 3D culture environment, liver progenitor cells continued to self-organize and mature into liver organoids. The growth factors or signal pathways involved in each differentiation stage are listed in the figure. BMP: Bone morphogenetic protein; DEX: Dexamethasone; ESCs: Embryonic stem cells; FGF: Fibroblast growth factor; HGF: Hepatocyte growth factor; iPSCs: Induced pluripotent stem cells; OSM: Oncostatin-M; PKA: Protein kinase A; PSCs: Pluripotent stem cells; PSC-LOs: Pluripotent stem cells derived liver organoids; RA: Retinoic acid; TGF-β: Transforming growth factor-beta; WNT: Wingless/integrated.

25% mTeSR™ medium and cholesterol mixture to the differentiation medium, resulting in the generation of liver organoids containing 60% ALB⁺ HLCs and approximately 30% CK19⁺ CLCs. Based on a similar approach, Ouchi *et al*^[33] developed a method for generating liver organoids through hiPSCs or hESCs differentiation. They first differentiated hPSCs into foregut endodermal cells, then embedded the cells into the Matrigel, further adding retinoic acid to generate liver organoids containing multiple cellular components. Single-cell RNA sequencing (scRNA-seq) analysis showed that the liver organoids they generated contained HLCs, CLCs, Kupffer cell-like cells, and hepatic stellate cells (HSCs).^[33] Many other studies

have also utilized hPSCs to generate liver organoids. These hPSC-LOs contain one or more cell types, which self-organize to form structural units similar to those in the liver,^[20–26] closely resembling liver structure and function.

Application of PSC-LOs

As mentioned previously, hPSC-LOs can simulate the structure and function of the human liver. Consequently, they have been extensively used in study on human liver development, liver disease modeling, drug development, personalized medicine, and liver regenerative medicine [Figure 2]

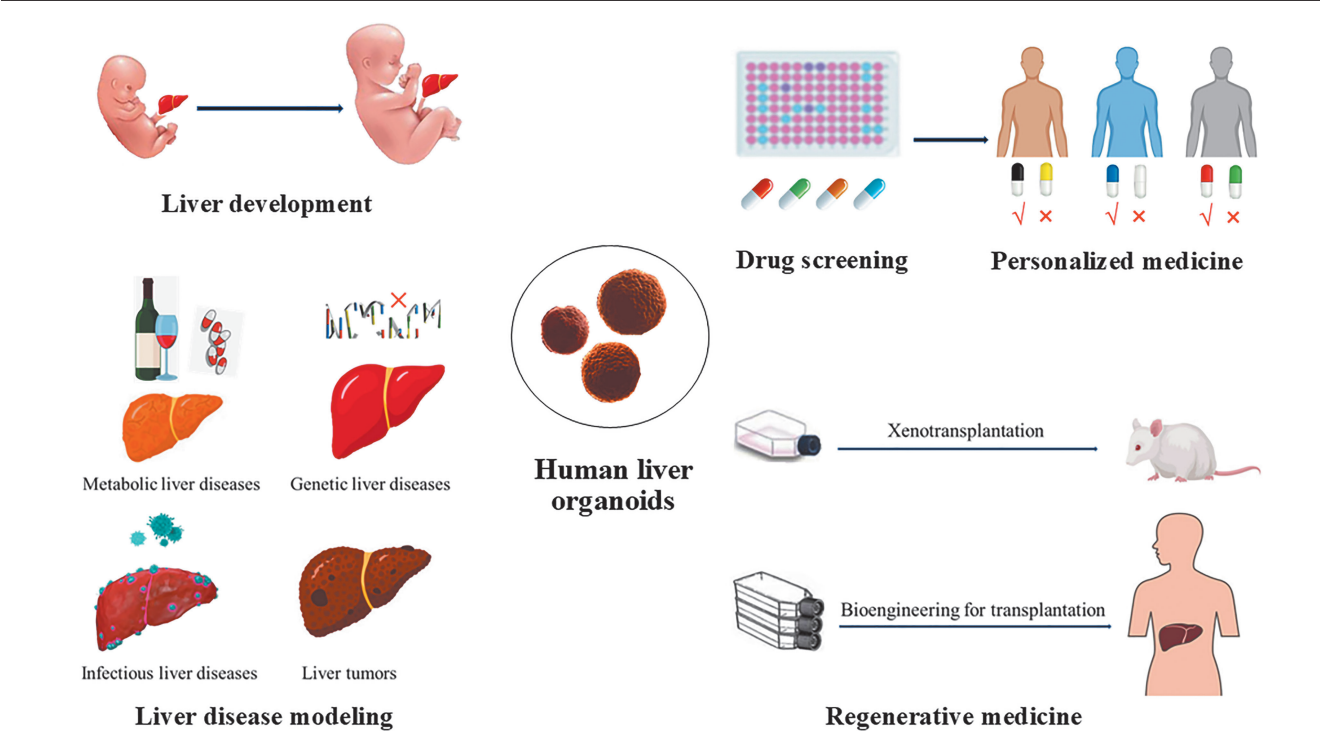


Figure 2: Application of PSC-LOs. Currently, PSC-LOs have been widely applied in the study of human liver development, and modeling of liver diseases, and have great potential in drug development, personalized medicine, and liver regenerative medicine. PSC-LOs: Pluripotent stem cells derived liver organoids.

Research on liver development

Development of the human liver is a complex process regulated by multiple signaling pathways. Although animal models have provided insights into liver development mechanisms, species differences limit their applicability to humans.^[34] Human ESCs and iPSCs were successfully differentiated into hepatocyte-like cells.^[28] However, early differentiation protocols were conducted under 2D conditions and lacked the 3D context required for proper liver tissue formation. Liver development involves hepatocyte generation, proliferation, and differentiation from the posterior foregut endoderm into hepatocytes and bile duct epithelial cells as well as the formation of mesenchymal cells from the mesoderm, which differentiate into liver fibroblasts and stellate cells. It is challenging to replicate these processes in 2D cultures.

The 3D cultured liver organoids leverage the self-organizing ability of stem and progenitor cells to simulate liver development *in vitro*. Takebe *et al*^[8] established LBs with 3D structures. When transplanted into mice, these LBs develop into liver tissues with mature characteristics, forming connections with the host circulatory system, suggesting that functional blood vessel formation triggers hPSC-derived LBs maturation. Interactions between cells and the matrix activate the fibroblast growth factor (FGF) and BMP pathways, which are essential for liver development.^[8,35] A study of liver organoids established by Asai *et al*^[19] showed that paracrine factors secreted by mesenchymal and endothelial cells, such as human growth factor (HGF), angiotensinogen (ANG), α -2 macroglobulin (A2M), and plasminogen (PLG), induce the formation of liver organoids, with significant differentiation and maturation observed after transplantation into immunodeficient mice. Koui *et al*^[36] generated liver organoids containing sinusoidal endothelial cells (LSECs) and HSCs from hiPSCs, confirming that the TGF- β and Rho signaling pathways regulate the proliferation and maturation of LSECs and HSCs progenitor cells. The establishment of these hPSC-LOs has enhanced our understanding of liver development.

Disease models

HPSC-LOs can be used to establish liver disease models for both genetic and acquired diseases. These models have advanced our understanding of liver disease mechanisms and have assisted in the development of effective treatments.

Genetic diseases

Alagille syndrome (ALGS) is an autosomal dominant inherited liver disease primarily caused by mutations in the *JAG1* gene (encoding the NOTCH ligand Jagged1).^[37] Changes in the NOTCH signal lead to impaired differentiation of the bile ducts, resulting in bile duct deficiency and chronic cholestasis in the liver.^[37] Guan *et al*^[31] generated liver organoids using hiPSCs derived from ALGS patients. Compared to control liver organoids, liver organoids derived from patients with ALGS showed impaired development of bile duct cells and bile duct structures.^[31] In addition, they introduced *JAG1* mutations into control

hiPSCs using CRISPR-associated protein 9 (Cas9) genome editing, obtaining liver organoids with phenotypes similar to those of hiPSC-LOs from ALGS patients.^[31]

Citrullinemia type 1 (CTLN1) is an inherited liver metabolic disorder caused by mutations in argininosuccinate synthase (ASS) gene, which leads to impaired urea circulation in the body.^[38] Akbari *et al*^[39] generated liver organoids from CTLN1 patient-specific iPSCs. Ammonia accumulation was observed in these iPSC-LOs, similar to the CTLN1 phenotype. In addition, the overexpression of *ASS1* in organoids can partially salvage the CTLN1 phenotype, which also proves that gene manipulation can be performed at the organoid level.^[39]

Glycogen storage disease type 1a (GSD1a), an autosomal recessive disease, is caused by defects in the glucose-6-phosphatase- α .^[40] Mun *et al*^[41] developed proliferative and functional liver organoids from iPSCs of patients with GSD1a. These PSC-LOs maintained disease-specific phenotypes, such as higher lipid and glycogen accumulation in liver organoid cells and lactate secretion into the culture media, consistent with GSD1a pathology.

Successful optimization of liver organoid culture technology has led to the establishment of models for various rare genetic liver diseases, highlighting the potential of PSC-LOs in genetic liver disease research.

Metabolic liver disease

Hepatocellular steatosis, commonly known as fatty liver disease, has become one of the most prevalent chronic liver diseases globally, primarily owing to the increasing rate of type 2 diabetes and obesity. Severe steatosis can develop to steatohepatitis.^[42]

Fatty Liver and Steatohepatitis: Mun *et al*^[43] established a liver organoid model derived from hiPSCs, and treated it with oleic acid (OA) and palmitic acid to mimic liver steatosis. After 4 days of treatment, lipid droplets were observed in hiPSC-LOs. Similarly, Ouchi *et al*^[33] constructed liver organoids containing HLCs, CLCs, Kupffer cells, and HSCs derived from hiPSCs. The treatment of hiPSC-LOs with OA led to dose-dependent lipid accumulation and the appearance of inflammatory and fibrotic phenotypes. Ramli *et al*^[44] established similar liver organoids to simulate non-alcoholic steatohepatitis (NASH). Co-incubation with free fatty acids results in gene expression changes and structural alteration characteristics of NASH, such as bile canaliculus decay and ductular reactions.

Acetaminophen-Induced Hepatotoxicity: Jiang *et al*^[23] used liver organoids derived from hPSC-derived foregut stem cells in a micropatterned agarose scaffold to evaluate the hepatotoxicity of acetaminophen (APAP). They observed APAP-induced upregulation of fibrotic markers (COL1A1, VIM) and pro-inflammatory cytokines (TNF- α , IL-8) in the organoids.

Alcoholic Liver Disease (ALD): Wang *et al*^[30] co-cultured human embryonic stem cell-derived organoids with human fetal liver mesenchymal cells to create an *in vitro*

ALD model. After ethanol treatment, the organoids exhibited symptoms of ALD, including steatosis, oxidative stress, fibrosis, and inflammation. Recently, we generated hiPSC-LOs and established an ALD model by treating liver organoids with alcohol. This model recapitulates the key pathological features of ALD, such as mitochondrial damage, elevated cellular reactive oxygen species (ROS) levels, fatty liver, and hepatocyte necrosis.^[45]

Infectious liver diseases

Viral hepatitis is among the most common and harmful infectious liver diseases, leading to approximately 1.4 million deaths annually worldwide, with approximately 90% attributable to hepatitis B and hepatitis C.^[46] Current research primarily utilizes human liver cancer cell lines and humanized mouse models, which have significant limitations in replicating the complex pathological processes of viral liver infections in humans.^[47] HiPSC-LOs can replicate the physiological structure and cellular polarity of the liver, while preserving its inherent immune response. Thus hiPSC-LOs are a promising model for the study of viral hepatitis. For instance, Nie *et al*^[48] established an HBV infection model using liver organoids containing multiple cellular components in a 3D micro-porous system, in which HBV could replicate and spread between cells for a long time. Organoids exhibit phenotypes such as liver dysfunction, appearance of liver injury markers, and changes in liver structure after infection with HBV.^[48] Compared with liver cells in a 2D culture system, this culture system is more susceptible to HBV infection and more accurately reproduces the host-virus interaction.^[48] Despite these advancements, PSC-LOs have not been widely used to model infectious liver diseases, primarily because of their low maturity. Therefore, the liver organoids currently used to study infectious liver diseases are mainly derived from adult stem cells.^[49,50] With the advancement of organoid culture technology and the increasing maturity of organoids, PSC-LOs are expected to demonstrate greater value in the research and treatment of infectious liver diseases.^[51]

Liver tumors

The advent of 3D culture technology for organoids has significantly impacted cancer research, particularly with the development of organoids from tumor tissues and cells, known as tumor organoids.^[52] Several studies have reported the use of gene editing techniques to introduce oncogene activation mutations into primary normal organoids, thereby driving the generation of tumor organoids.^[53,54] PSC-LOs can contain various liver cell components, reproducing the inherent structure and physiological functions of the liver *in vitro*. They can also be gene-edited using CRISPR-Cas9 technology, providing a new approach for establishing liver tumor models.^[55,56] However, there are currently no reports yet of oncogene-driven tumor organogenesis using PSC-LOs. This will also be the next valuable research field.

In a previous study, Artegiani *et al*^[57] used CRISPR-Cas9 to delete *BAP1* in human ductal liver organoids. When

combined with other cholangiocarcinoma associated mutations, this led to tumor formation in mice. In another study, Sun *et al*^[58] established liver organoids with liver structure and function using human liver cells directly reprogrammed from immortalized umbilical cord fibroblasts. They genetically engineered the organoids to simulate the initial alterations in human liver cancer and found that c-Myc induced mitochondrion–endoplasmic reticulum over-coupling promotes the occurrence of hepatocellular carcinoma, which seems to be a target of preventive treatment.^[58] Guan *et al*^[59] differentiated hiPSCs into 3D human liver organoids and discovered, through single-cell RNA-sequencing and metabolomics, that early hepatoblasts were sensitive to meclizine. The study of organoids binding omics profiling analysis is expected to become another new prospective direction for the application of organoids in cancer research.

Currently, patient-derived tumor organoids (PDTOs) are widely used organoid models for liver tumor research, particularly for tumor drug screening and efficacy evaluation. However, they do not fully replicate tumor initiation and transformation processes. The combination of PSC-LOs and gene editing technology has enormous potential in studying the occurrence, development, and transformation of liver tumors, as well as the key signaling pathways and metabolic processes involved in tumor development. It is foreseeable that PSC-LOs will play an important role in research on liver tumors in the future.

Some key LOs disease models are presented in the Table 1.^[23,30,31,33,39,41,43–45,48,58,60–67] As research on liver organoid models advances, it is anticipated that more sophisticated models will emerge, capable of more accurately reflecting disease physiology in the future.

Drug development and personalized medicine

Drug screening is crucial for the development of treatments for liver diseases and has traditionally relied on animal models and cell cultures.^[68] However, genetic differences between species often lead to inaccurate predictions of human drug effects in animal models and are not suitable for high-throughput drug screening.^[4,69] PHHs, considered the “gold standard” for drug efficacy and toxicity evaluation,^[70] have limitations in terms of availability, proliferation *in vitro*, and relevance to non-parenchymal liver cells, which are crucial in many liver diseases.^[2,71,72] As an alternative, researchers have developed a drug screening model based on hPSCs-derived liver cells or organoids to evaluate the efficacy and toxicity of potential candidate drugs and predict the likelihood of potential drugs causing serious side effects.^[30,33,73–75] For example, some studies have used hPSCs-derived hepatocytes for large-scale drug screening and toxicity testing, and the results showed that their ability to predict drugs is comparable to that of PHHs.^[73,74] Liver organoids generated in a 3D culture environment close to the physiological microenvironment can accurately reflect the liver’s response to drugs because of their stereo-structures. The HPSC-LOs have also been used for large-scale drug screening and toxicity testing.^[76–78] Some hPSC-LOs are more sensitive to drug screening and toxicity testing for

Table 1: Some liver disease models established by hPSC-LOs.

Disease models	Cell sources	Description	References
Genetic disease			
ALGS	iPSCs from health controls and ALGS patients	Liver organoids were generated from hiPSCs of ALGS patients. Introduction of <i>JAG1</i> mutations into normal hiPSCs can induce the phenotypes of hiPSC-LOs from ALGS patients.	[31]
CTLN1	iPSCs from CTLN1 patients and a healthy donor	CTLN1 patient-derived iPSC-LOs recapitulate disease phenotypes (ammonia accumulation, reduced ureagenesis), partially rescued by <i>ASS1</i> overexpression.	[39]
GSD1a	iPSCs from GSD1a patients	GSD1a patient-derived iPSC-LOs successfully modeled key disease phenotypes, including pathological higher lipid and glycogen accumulation characteristic of GSD1a.	[41]
WD	iPSCs from health controls and WD patients	WD patient-derived iPSC-LOs showed exacerbated OA-induced steatosis and fibrosis compared to non-Wolman derived LOs.	[33]
CF	iPSCs from a CF patient	CF patient iPSC-derived cholangiocyte organoids showed secretin-induced size increase, while somatostatin and octreotide caused size reduction, modeling CF-related secretory responses.	[60]
CF	iPSCs from CF patients	Generated cholangiocyte organoids from normal and CF iPSCs, revealing defective forskolin-induced cyst swelling in CF models that was rescued by CFTR modulators.	[61]
OTCD	iPSCs from an OTCD patient	Corrected an <i>OTCD</i> mutation in patient iPSCs via homologous directed repair, then differentiated them into hepatocyte-like cells organoids and characterized their phenotypes.	[62]
Metabolic liver disease			
Hepatic steatosis	iPSCs induced from human HFFs	HiPSC-derived liver organoids treated with fatty acids (OA/palmitate) for 3 days showed steatosis phenotypes: increased intracellular lipids and upregulated lipid metabolism metabolites.	[43]
Steatohepatitis	iPSCs and hESCs	ESC/iPSC-derived liver organoids that mimicked steatohepatitis (steatosis, inflammation, fibrosis) upon OA treatment were created.	[33]
NASH	hESCs and iPSCs	iPSC/ESC-derived liver organoids were developed, showing elevated lipid droplets, ROS, lipid peroxidation, and metabolic gene expression after 4-day FFA treatment.	[44]
ALD	hESCs and hFLMC	hEHO combined with hFLMC replicated ALD features (oxidative stress, steatosis, and fibrosis) upon ethanol treatment.	[30]
ALD	iPSCs	An ALD model using ethanol-treated hiPSC-derived liver organoids was established, recapitulating key ALD features (steatosis, mitochondrial damage, ROS elevation, and hepatocyte necrosis).	[45]
Liver fibrosis	hiPSCs	hPSC-derived liver organoids that exhibited APAP-triggered hepatotoxicity and fibrosis were developed by micropatterned agarose scaffolds.	[23]
Infectious liver diseases			
Hepatitis B	iPSCs	3D iPSC-derived liver organoid model was developed with persisting HBV replication and recapitulating liver dysfunction, gene expression downregulation, acute failure markers release, and liver ultrastructural changes.	[48]
Hepatitis C	iPSCs	iPSC-derived liver organoids were developed expressing HCV host factors and showing susceptibility to genotype 2a HCV infection.	[63]
Hepatitis B	iPSCs	iPSC-derived hepatocyte organoids were developed and constructed an <i>in vitro</i> HBV infection model by inoculating HBV particles in the system.	[64]
COVID-19	iPSCs	ACE2 expression in hPSC-LOs and their permissiveness to SARS-CoV-2 pseudo-entry virus infection.	[65]
COVID-19	iPSCs	SARS-CoV-2 susceptibility in PSC-LOs, with hepatocyte-like cells showing the strongest inflammatory response post-infection.	[66]
Sepsis-associated liver dysfunction	iPSCs	By co-culturing hiPSC-EMPs and hiPSC-LOs, Kupffer cell-containing organoids that exhibited sepsis-like liver dysfunction upon LPS/IFN- γ exposure were induced.	[67]
Liver tumors			
HCC and ICC	Immortalized umbilical cord fibroblasts	Functional liver organoids from reprogrammed umbilical fibroblasts was established and engineered to model the initial alterations in HCCs and ICCs.	[58]

ALD: Alcoholic liver disease; ALGS: Alagille syndrome; APAP: Acetaminophen; CF: Cystic fibrosis-associated liver disease; CFTR: Cystic fibrosis transmembrane conductance regulator; COVID-19: Corona virus disease 2019; CTLN1: Citrullinemia type 1; EMPs: Erythroid progenitor cells; FFA: Free fatty acid; GSD1a: Glycogen storage disease type 1a; HBV: Hepatitis B virus; HCC: Hepatocellular carcinoma; HCV: Hepatitis C virus; hEHO: Human ESC-derived liver organoids; HFFs: foreskin fibroblasts; hFLMC: Human fetal liver mesenchymal cells; ICC: Intra-Hepatic cholangiocarcinoma; IFN- γ : Interferon- γ ; iPSC: Induced pluripotent stem cells; LOs: Liver organoids; LPS: Lipopolysaccharide; NASH: Nonalcoholic steatohepatitis; OA: Oleic acid; ROS: Reactive oxygen species; OTCD: Ornithine transcarbamylase deficiency; SARS-CoV-2: Severe acute respiratory syndrome coronavirus 2; WD: Wolman disease.

drug-induced liver injury (DILI), inflammatory liver disease, liver fibrosis, and other liver diseases because of the

presence of non-parenchymal cells, such as hepatic stellate cells and Kupffer cells.^[23,30,33,75,79] Some hPSC-LOs

also have vascular structures that can simulate the process of drug delivery through blood vessels to various internal cells during drug screening and toxicity testing, making them closer to real-world *in vivo* situations.^[17,20] Given the genetic variability among patients, hPSCs-derived liver cells and organoids that retain patient-specific genetic information offer valuable tools for studying disease mechanisms, screening drugs, and exploring personalized treatment strategies.^[80] Tables 2 and 3 give some examples of toxicity testing and drug screening based on PSC-LOs, respectively.^[23,30,33,43–45,60,61,75–79,81–85]

However, in most of these studies, the scale of PSC-LOs used for drug screening was relatively small and the drug evaluation methods were different, which limited the mutual authentication of drug screening results in different studies. Therefore, it is necessary to use bioengineering technology to expand the scale of drug screening.^[81] At the same time, it is also necessary to combine the recently popular artificial intelligence (AI) technology to develop more automatic, intelligent, and scientific drug efficacy evaluation systems, accurately evaluate drug screening results, and even predict the test results of various new drug combinations.^[86,87]

Cell therapy and regenerative medicine

Orthotopic allogeneic liver transplantation is currently the only effective treatment for end-stage liver failure. However, there is a significant shortage of healthy liver tissue suitable for transplantation, meeting the needs of only about 10% of patients. Additionally, patients require lifelong immunosuppressive therapy post-transplantation, making this treatment impractical as a routine solution. To address these challenges, researchers have explored alternative strategies, such as xenograft liver transplantation, bioartificial liver, and liver cell transplantation.^[88] However, xenotransplants raise concerns about immune rejection and ethical issues, and bioartificial liver technology is not yet clinically viable.^[89] However, due to limited sources of liver cells, poor quality of isolated liver cells, and occasional occurrence of liver cell immune rejection^[90], liver cell transplantation cannot completely cure the disease and can only serve as a temporary solution while waiting for liver transplantation.^[91]

With the successful isolation of hESCs and the emergence of iPSCs, researchers have identified hPSC-derived liver cells as potential cell sources for transplantation^[92–94] Liver organoids, especially hiPSC-LOs, retain the genetic

Table 2: Hepatic toxicity assessment based on PSC-LOs.

Organoids sources	Drugs	Evaluation index	References
hiPSCs	Troglitazone; APAP; rotenone; trovafloxacin; levofloxacin.	Cell viability; reactive oxygen species; cellular glutathione content; nuclei structure; mitochondrial respiration.	[43]
hiPSCs	238 marketed drugs in hepatotoxicity library.	Cell viability.	[75]
hiPSCs	APAP	CYP450 activity.	[84]
hiPSCs	APAP	Cell viability.	[85]
hiPSCs	APAP; fialuridine; tenofovir; inarigivir soproxil.	Cell viability.	[82]
hESCs	Microplastics; bisphenol A.	Cytotoxicity; oxidative stress; inflammatory response; lipid accumulation.	[76]
hiPSCs and hESCs	Ticlopidine; flutamide; troglitazone; benzbromarone; diclofenac; APAP; valproic acid; nefazodone; ketoconazole; Ffluconazole; rosiglitazone; buspirone.	CYP450 activity; cytotoxicity.	[77]
hiPSCs	APAP; MTX; fasiglifam; fialuridine; TGF-β1; LPS.	ROS intensity; mitochondrial depolarization; lipid accumulation; ATP content; albumin secretion.	[78]
hiPSCs	APAP	Cell viability/cytotoxicity; expression of fibrotic and inflammatory genes; collagen1A1 protein.	[23]
hESCs and hiPSCs	Troglitazone	Cholestasis	[44]
hESCs and hFLMC	Ethanol	Dead/Live staining; fibrogenic genes; pro-collagen type I secretion; sirius red staining.	[30]
hiPSCs	Oleic acid; linoleic acid; stearic acid; palmitic acid.	Lipid accumulation; fibrosis phenotypes; stiffness of organoids.	[33]
hiPSCs	Parenteral nutrition.	Lipid accumulation; Organoid viability; CYP3A4 activity; Hepatocyte functional genes expression.	[79]
hiPSCs	Ethanol	Cytotoxicity; lipid accumulation; ROS levels; MMP.	[45]
hiPSCs	Secretin; somatostatin; octreotide.	Organoid diameter; cAMP levels.	[60]
hiPSCs	Sorafenib; tamoxifen.	Cell viability	[81]

APAP: Acetaminophen; ATP: Adenosine triphosphate; cAMP: Cyclic adenosine monophosphate; CYP450: Cytochrome P450; hESC: Human embryonic stem cell; HFFs: Foreskin fibroblasts; hFLMC: Human fetal liver mesenchymal cells; hiPSCs: Human induced pluripotent stem cells; LPS: Lipopolysaccharides; MMP: Mitochondrial membrane potential; MTX: Methotrexate; PSC-LOs: PSC-derived liver organoids; ROS: Reactive oxygen species; TGF-β1: Transforming growth factor-β1.

Table 3: Examples of drug validation and testing based on PSC-LOs.

Disease models	Cell sources	Drugs	Evaluation index	References
Liver fibrosis	hiPSCs	SD208; imatinib; cilofexor; silymarine.	Expression of fibrosis related genes; collagen I immunofluorescence staining.	[78]
WD	hiPSCs/hESCs	D-penicillamine; bathocuproine-disulfonic acid; trientine hydrochloride; DL- α -Tocopherol acetate.	Cell viability	[77]
ARPKD liver disease	hiPSCs	Crenolanib; sunitinib; imatinib	Fibrosis scores	[83]
WD	hiPSCs from WD patients	Obeticholic acid; recombinant FGF19; recombinant lysosomal acid lipase; magnesium.	Lipid accumulation; fibrosis phenotypes; Stiffness of organoids; survival rate.	[33]
Liver steatosis	hiPSCs	Metformin; etomoxir; L-carnitine; everolimus; scriptaid; tacedinaline; KU- 0063794.	Lipid accumulation; fatty acid metabolism genes expression.	[43]
ALD	hiPSCs	L-22; metadoxine; N-acetylcysteine.	Cell viability	[45]
CF liver disease	hiPSCs from CF patients	VX809; VX809 plus CFTR inhibitor 172.	Organoid diameter; CFTR activity.	[60]
CF liver disease	hiPSCs from CF patients	Forskolin; IBMX and the CFTR potentiator VX-770.	Degree of organoids swelling.	[61]

ALD: Alcoholic liver disease; ARPKD: Autosomal Recessive Polycystic Kidney Disease; CF: Cystic fibrosis; CFTR: Cystic fibrosis transmembrane regulator; FGF19: Fibroblast growth factor 19; hESCs: Human embryonic stem cells; hiPSCs: Human induced pluripotent stem cells; IBMX: 3-isobutyl-1-methylxanthine; IL-22: Interleukin 22; WD: Wolman disease.

information of patients and can expand, differentiate, mature, and reproduce the structure and function of the human liver *in vitro*, making them an ideal alternative source for liver transplantation. Takebe *et al*^[8] transplanted human iPSC-derived LBs into mouse skull windows and found that human iPSC LBs formed a liver cord-like structure, with protein expression profiles and liver-specific functions similar to LBs in human fetal liver, and may differentiate into the mature liver. Further transplantation of human iPSC-derived LBs into an immunodeficient mouse liver failure model can significantly improve survival rate and prolong survival time.^[8,17] Rashidi *et al*^[95] generated 3D hiPSC-LOs, which can be cultured *in vitro* for up to one year and maintain a stable phenotype. After binding with a polycaprolactone scaffold, they were transplanted into two tyrosinemia mouse models through both intraperitoneal and subcutaneous methods. The results showed that these hiPSC-LOs secreted human albumin after survival and compensated for damaged liver function in mice. In two other studies, researchers transplanted iPSC-LOs into rats and piglets through the portal vein, demonstrating the safety of organoid transplantation through the portal vein.^[96,97] Overall, these advancements suggest that liver organoids could become a viable alternative for liver transplantation and cell therapy in the future, providing new hope for patients with liver diseases.

Limitations and Future Directions of PSC-LOs

HPSC-LOs have demonstrated potential in areas such as disease modeling, drug development, and regenerative medicine.^[51,55,98] However, it is worth mentioning that the field of organoid research is still in its early stages. Currently, organoids predominantly function as technical tools and exhibit substantial differences from the native target organs. Several scientific and technological

limitations hinder the application of human PSC-LOs from the workbench to bedside.^[51] Currently, liver organoids face several key challenges [Figure 3].

Maturity and functionality

The low maturity of PSC-LOs has been a significant limitation, most exhibiting characteristics similar to fetal liver cells, including positive markers for liver progenitor and early liver cell-specific markers.^[23] This immaturity impacts their application in disease modeling and drug development. Researchers have been using various methods to promote the efficiency of liver differentiation and improve the maturity of liver organoids. For example, Mun *et al*^[99] found that treatment of liver organoids with microbial short-chain fatty acids (SCFAs) increased CYP3A4 activity, albumin secretion, and gene expression, resulting in significant improvements in liver maturation of liver organoids. Additionally, Zahmatkesh *et al*^[100] showed that the incorporation of cell-sized microparticles (MPs) derived from liver extracellular matrix (ECM) into their cultured bioengineered liver organoids (BLOs) improved the maturation of hepatocytes inside the BLOs. Furthermore, combining organoid technology with new bioengineering tools, such as microfluidics, micropatterning, biochips and AI technology, and biomaterial scaffolds, to precisely control the cell culture microenvironment and improve the functional maturity of human PSC-LOs, have achieved great process.^[20,51,81,101–104]

Size and scalability

A significant challenge for PSC-LOs is the gap in size and scale compared with normal liver tissues, making them unsuitable for clinical transplantation applications.^[18,51] The current size of liver organoids is mainly on the

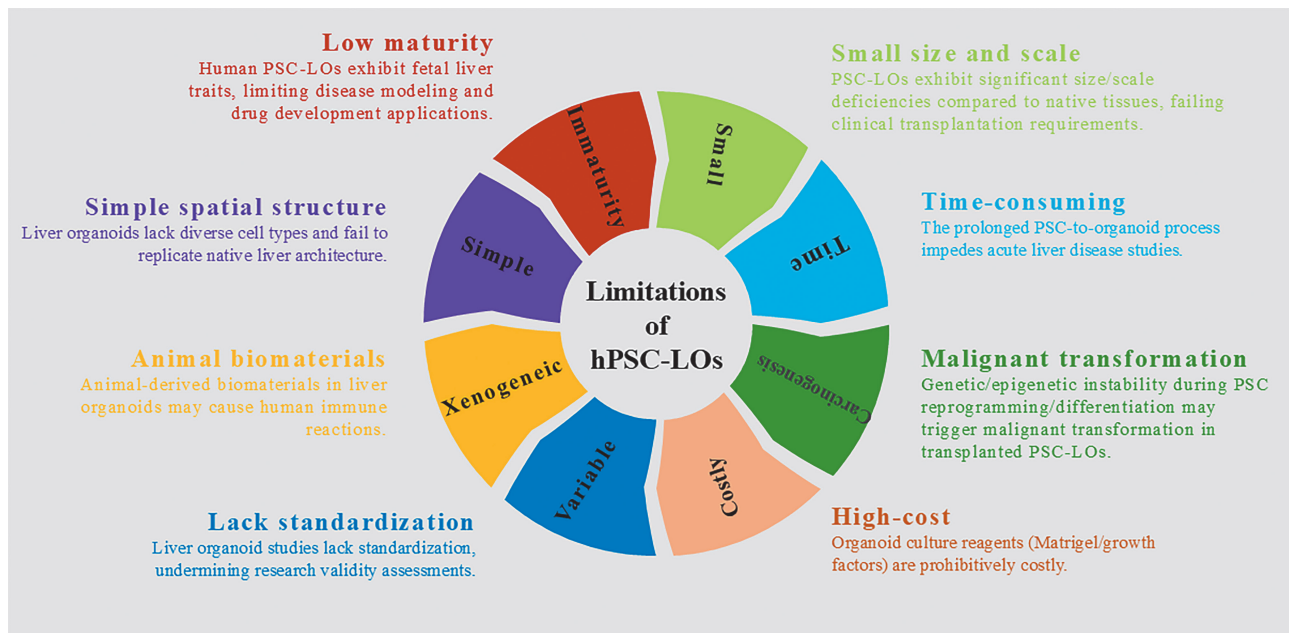


Figure 3: Limitations of PSC-LOs. Liver organoid research remains in its early developmental stage, with multifaceted limitations impeding clinical translation of human PSC-LOs. hPSCs: Human pluripotent stem cells; PSC-LOs: Pluripotent stem cells derived liver organoids.

micron-sized scale, primarily owing to limitations such as lack of vascular system, imbalanced nutrient permeation, insufficient oxygen supply in the core area, and difficulty in metabolite excretion.^[105] For liver transplantation or cell therapy, the required number of cells is at least one-tenth of the liver mass, which is approximately 1×10^9 cells.^[106,107] Although PSCs have a strong amplification ability and some PSC-LOs also have the ability for long-term passage amplification,^[30,39] most liver organoids generated in laboratories currently cannot meet the requirements for transplantation. Therefore, to meet the clinical needs of cell quantity, it is necessary to continue optimizing organoid culture techniques to promote cell proliferation while combining bioengineering technology and tools such as microfluidics, micropatterning, and biochips to expand the cultivation scale.^[17,81,108]

Limited cellular composition and simple spatial structure

Currently, the cellular composition and tissue structure of liver organoids are not fully representative of actual liver tissue. Ideally, PSC-LOs should include all types of liver cells in their natural ratios. However, no existing liver organoids fully achieve this. Some PSC-LOs do contain multiple liver cell types, such as hepatocytes, cholangiocytes, Kupffer cells, and hepatic stellate cells. Recent studies have also reported vascularized liver organoids containing endothelial cells,^[20,21] and have detected the expression of neuronal lineage markers.^[20] However, these PSC-LOs from different laboratories exhibit heterogeneity in cell composition, morphology, and function. Additionally, the tissue structure of these PSC-LOs is still too simple and far from the basic liver lobular structure. It cannot simulate the structure and function of different liver regions. Techniques such as organoid co-culture systems, biochips, and 3D bioprinting are expected to achieve PSC-LOs

with diverse cell types, complex structures, and comprehensive functions. For instance, co-culturing organoids with endothelial cells can improve vascularization,^[21] while incorporating immune cells can better simulate the immune microenvironment.^[50] 3D bioprinting not only offers a cost-effective way to enhance vascularization,^[109] which is expected to overcome the limitations of delivering nutrients and oxygen to organoids, but also allows for the standardization and quality control of organoid production to minimize human intervention.^[86] The combination of biochip technology and organoid technology enables the precise optimization of culture conditions in organoid culture systems, maintains a constant balance of nutrients, growth factors, and metabolites, and facilitates the accurate simulation of native liver tissue *in vitro*.^[104,110–113]

Malignant transformation

Malignant transformation of iPSC-LOs after transplantation is also a noteworthy issue.^[18] This is mainly due to genetic and epigenetic instabilities introduced during the reprogramming and differentiation of human iPSCs. Reprogramming protocols using integrative vectors, such as lentiviruses and retroviruses, can randomly integrate viral genes into the iPSCs genome, potentially leading to tumorigenesis.^[114] Studies have shown that duplication of oncogenes tends to occur during the long-term passage of human iPSCs, the duplication of oncogenes tends to accumulate.^[115] Therefore, it is recommended to use non-integrative vectors in reprogramming protocols and to select PSC lines with fewer passages and stable phenotypes for differentiation. Moreover, optimizing reprogramming protocols and culture conditions while monitoring genetic and epigenetic variations in cells throughout the culture process is essential to ensure

genomic integrity.^[116] Extensive investigations in animal models are necessary to address this issue before PSC-LOs can be safely used for clinical transplantation.^[18]

Side effects of animal biomaterials

Another concern is the use of animal-derived materials, such as Matrigel and animal serum, in current liver organoid differentiation protocols. These materials can cause adverse reactions in the human body.^[18,51] To address this issue, the development and validation of new implantable animal-free biomaterials are crucial for creating clinically applicable organoids. For example, Sekine and colleagues cultured human iPSCs-derived LBs using chemically defined, animal-origin-free media as early as 2020.^[117] Alternative xenofree bioengineered materials such as polyethylene glycol hydrogel,^[118] polyisocyanopeptides based hydrogels,^[119] nanofibrillar hydrogel with controllable stiffness,^[120] inverted colloidal crystal scaffolds with type I collagen,^[63] and hyaluronic acid-based hydro-scaffold (Biomimesy)^[121] have been developed to replace Matrigel for the safe application of liver organoids in regenerative medicine.^[122] However, these strategies increase the cost of liver organoids.

Time-consuming and high cost

The generation of PSC-LOs takes longer compared to those from adult stem cells. The reprogramming process of iPSCs further lengthens the modeling cycle, which is particularly challenging for researching acute liver diseases where timely intervention is crucial.^[123] Therefore, researchers have proposed a strategy for allogeneic cell transplantation. For example, HLA gene editing mediated by CRISPR-Cas9 can enhance the immune compatibility of hiPSCs.^[124,125] The application of these non-immune responsive allogeneic hiPSC-derived organoids can omit the induction process of patient iPSC, thereby reducing the modeling time. In addition, directly differentiating the patient's mesenchymal stem cells (such as adipose stromal/stem cells)^[126] or directly reprogramming the patient's fibroblasts^[127] into liver organoids can also save time in model construction. The cost of materials like Matrigel, media supplements, and growth factors required for culturing organoids is another significant barrier to large-scale application. Developing low-cost bioactive molecules and optimizing differentiation protocols could help reduce these costs.

Lack of standardization

Finally, a notable limitation in the field of liver organoid research is lack of standardization, particularly concerning the establishment and quality control of organoids. Variations in the morphology, composition, and maturity of the liver organoids generated by different laboratories using different protocols significantly hinder their specificity and reproducibility. Although the liver organoid culture technology continues to advance, establishing a universally accepted standard protocol for generating these organoids remains challenging. Therefore unified evaluation standards or guidelines are urgently needed

for assessing liver organoids based on their structural and functional similarities to primary liver tissue.^[51,128,129] Sang *et al.*^[129] evaluated liver organoids based on three main aspects: gene expression profiling, cell phenotypes, and liver-specific functions. Techniques such as single-cell profiling, gene expression profiling, flow cytometry, AI-based phenotype analysis, and other functional assays may serve as advanced approaches for establishing the standardization in liver organoid research.^[87,128,129] Additionally, as an emerging technology, liver organoid research lacks authoritative guidelines or consensus of ethical governance. For instance, technical specifications of PSCs gene editing, consent processes for organoid biobank donations, and ethical review systems for future clinical transplantation all require the establishment of standardized frameworks.

Conclusion

Liver organoid technology has undergone significant advancements over the past decade, demonstrating its transformative potential in areas such as tissue development, disease modeling, regenerative medicine, and drug development. However, significant limitations still require resolution. In the future, interdisciplinary cooperation in biochemistry, cell biology, bioengineering, biomaterial engineering, biomechanics, and clinical medicine is needed to advance the PSC-LOs technology toward clinical application standards.

Funding

This work was supported by grants from the Fund of Shanxi "1331 Project" Key Subjects Construction (No. 1331KSC), the Fund of Science and Technology Innovation Plan for Colleges and Universities from Education Department of Shanxi Province (No. 2019L0423), the Key R&D Program of Shanxi Province (International Cooperation, No. 201903D421023), and the Shanxi Basic Research Program (Nos. 20210302124406 and 202103021223227).

Conflicts of interest

None.

References

1. Yuan E, Ye Z, Song B. Imaging-based deep learning in liver diseases. *Chin Med J* 2022;135:1325–1327. doi: 10.1097/CM9.0000000000002199.
2. Xiang C, Du Y, Meng G, Soon Yi L, Sun S, Song N, *et al.* Long-term functional maintenance of primary human hepatocytes in vitro. *Science* 2019;364:399–402. doi: 10.1126/science.aau7307.
3. Horbach S, Halffman W. The ghosts of HeLa: How cell line misidentification contaminates the scientific literature. *PLoS One* 2017;12:e0186281. doi: 10.1371/journal.pone.0186281.
4. Cheon DJ, Orsulic S. Mouse models of cancer. *Annu Rev Pathol* 2011;6:95–119. doi: 10.1146/annurev.pathol.3.121806.154244.
5. Harrison SP, Baumgarten SF, Verma R, Lunov O, Dejneka A, Sullivan GJ. Liver Organoids: Recent Developments, Limitations and Potential. *Front Med (Lausanne)* 2021;8:574047. doi: 10.3389/fmed.2021.574047.
6. Prior N, Inacio P, Huch M. Liver organoids: From basic research to therapeutic applications. *Gut* 2019;68:2228–2237. doi: 10.1136/gutjnl-2019-319256.
7. Sato T, Vries RG, Snippert HJ, van de Wetering M, Barker N, Stange DE, *et al.* Single Lgr5 stem cells build crypt-villus structures

- in vitro without a mesenchymal niche. *Nature* 2009;459:262–265. doi: 10.1038/nature07935.
8. Takebe T, Sekine K, Enomura M, Koike H, Kimura M, Ogaeri T, *et al.* Vascularized and functional human liver from an iPSC-derived organ bud transplant. *Nature* 2013;499:481–484. doi: 10.1038/nature12271.
 9. Lancaster MA, Renner M, Martin CA, Wenzel D, Bicknell LS, Hurler ME, *et al.* Cerebral organoids model human brain development and microcephaly. *Nature* 2013;501:373–379. doi: 10.1038/nature12517.
 10. McCracken KW, Catá EM, Crawford CM, Sinagoga KL, Schumacher M, Rockich BE, *et al.* Modelling human development and disease in pluripotent stem-cell-derived gastric organoids. *Nature* 2014;516:400–404. doi: 10.1038/nature13863.
 11. Takasato M, Er PX, Chiu HS, Maier B, Baillie GJ, Ferguson C, *et al.* Kidney organoids from human iPSCs contain multiple lineages and model human nephrogenesis. *Nature* 2015;526:564–568. doi: 10.1038/nature15695.
 12. Völkner M, Zschätzsch M, Rostovskaya M, Overall RW, Busskamp V, Anastassiadis K, *et al.* Retinal organoids from pluripotent stem cells efficiently recapitulate retinogenesis. *Stem Cell Reports* 2016;6:525–538. doi: 10.1016/j.stemcr.2016.03.001.
 13. Dye BR, Hill DR, Ferguson MA, Tsai YH, Nagy MS, Dyal R, *et al.* In vitro generation of human pluripotent stem cell derived lung organoids. *Elife* 2015;4:e05098. doi: 10.7554/eLife.05098.
 14. Huch M, Koo BK. Modeling mouse and human development using organoid cultures. *Development* 2015;142:3113–3125. doi: 10.1242/dev.118570.
 15. Huch M, Dorrell C, Boj SF, van Es JH, Li VS, van de Wetering M, *et al.* In vitro expansion of single Lgr5+ liver stem cells induced by Wnt-driven regeneration. *Nature* 2013;494:247–250. doi: 10.1038/nature11826.
 16. Huch M, Gehart H, van Boxtel R, Hamer K, Blokzijl F, Verstegen MM, *et al.* Long-term culture of genome-stable bipotent stem cells from adult human liver. *Cell* 2015;160:299–312. doi: 10.1016/j.cell.2014.11.050.
 17. Takebe T, Sekine K, Kimura M, Yoshizawa E, Ayano S, Koido M, *et al.* Massive and reproducible production of liver buds entirely from human pluripotent stem cells. *Cell Rep* 2017;21:2661–2670. doi: 10.1016/j.celrep.2017.11.005.
 18. Sakabe K, Takebe T, Asai A. Organoid medicine in hepatology. *Clin Liver Dis (Hoboken)* 2020;15:3–8. doi: 10.1002/cld.855.
 19. Asai A, Aihara E, Watson C, Mourya R, Mizuochi T, Shivakumar P, *et al.* Paracrine signals regulate human liver organoid maturation from induced pluripotent stem cells. *Development* 2017;144:1056–1064. doi: 10.1242/dev.142794.
 20. Harrison SP, Siller R, Tanaka Y, Chollet ME, de la Morena-Barrio ME, Xiang Y, *et al.* Scalable production of tissue-like vascularized liver organoids from human PSCs. *Exp Mol Med* 2023;55:2005–2024. doi: 10.1038/s12276-023-01074-1.
 21. Kim HJ, Kim G, Chi KY, Kim H, Jang YJ, Jo S, *et al.* Generation of multilineage liver organoids with luminal vasculature and bile ducts from human pluripotent stem cells via modulation of Notch signaling. *Stem Cell Res Ther* 2023;14:19. doi: 10.1186/s13287-023-03235-5.
 22. Kamishibahara Y, Okamoto S, Ohkuma T, Taniguchi H. Stabilized generation of human iPSC-derived liver organoids using a modified coating approach. *Biol Methods Protoc* 2023;8:bpac034. doi: 10.1093/biomed/bpac034.
 23. Jiang S, Xu F, Jin M, Wang Z, Xu X, Zhou Y, *et al.* Development of a high-throughput micropatterned agarose scaffold for consistent and reproducible hPSC-derived liver organoids. *Biofabrication* 2022;15. doi: 10.1088/1758-5090/ac933c.
 24. Raggi C, Sella S, McCallum MA, Paganelli M. Generation of Complex Syngeneic Liver Organoids from Induced Pluripotent Stem Cells to Model Human Liver Pathophysiology. *Curr Protoc* 2022;2:e389. doi: 10.1002/cpz1.389.
 25. Pettinato G, Perelman LT, Fisher RA. Development of a Scalable Three-Dimensional Culture of Human Induced Pluripotent Stem Cells-Derived Liver Organoids. *Methods Mol Biol* 2022;2455:131–147. doi: 10.1007/978-1-0716-2128-8_12.
 26. Maepa SW, Marakalala MJ, Ndlovu H. Generation of Multicellular 3D Liver Organoids From Induced Pluripotent Stem Cells as a Tool for Modelling Liver Diseases. *Bio Protoc* 2024;14:e5042. doi: 10.21769/BioProtoc.5042.
 27. Ang LT, Tan A, Autio ML, Goh SH, Choo SH, Lee KL, *et al.* A Roadmap for Human Liver Differentiation from Pluripotent Stem Cells. *Cell Rep* 2018;22:2190–2205. doi: 10.1016/j.celrep.2018.01.087.
 28. Si-Tayeb K, Noto FK, Nagaoka M, Li J, Battle MA, Duris C, *et al.* Highly efficient generation of human hepatocyte-like cells from induced pluripotent stem cells. *Hepatology* 2010;51:297–305. doi: 10.1002/hep.23354.
 29. Pettinato G, Lehoux S, Ramanathan R, Salem MM, He LX, Muse O, *et al.* Generation of fully functional hepatocyte-like organoids from human induced pluripotent stem cells mixed with Endothelial Cells. *Sci Rep* 2019;9:8920. doi: 10.1038/s41598-019-45514-3.
 30. Wang S, Wang X, Tan Z, Su Y, Liu J, Chang M, *et al.* Human ESC-derived expandable hepatic organoids enable therapeutic liver repopulation and pathophysiological modeling of alcoholic liver injury. *Cell Res* 2019;29:1009–1026. doi: 10.1038/s41422-019-0242-8.
 31. Guan Y, Xu D, Garfin PM, Ehmer U, Hurwitz M, Enns G, *et al.* Human hepatic organoids for the analysis of human genetic diseases. *JCI Insight* 2017;2:e94954. doi: 10.1172/jci.insight.94954.
 32. Wu F, Wu D, Ren Y, Huang Y, Feng B, Zhao N, *et al.* Generation of hepatobiliary organoids from human induced pluripotent stem cells. *J Hepatol* 2019;70:1145–1158. doi: 10.1016/j.jhep.2018.12.028.
 33. Ouchi R, Togo S, Kimura M, Shinozawa T, Koido M, Koike H, *et al.* Modeling Steatohepatitis in Humans with Pluripotent Stem Cell-Derived Organoids. *Cell Metab* 2019;30:374–384.e6. doi: 10.1016/j.cmet.2019.05.007.
 34. Shinozawa T, Yoshikawa HY, Takebe T. Reverse engineering liver buds through self-driven condensation and organization towards medical application. *Dev Biol* 2016;420:221–229. doi: 10.1016/j.ydbio.2016.06.036.
 35. Takebe T, Enomura M, Yoshizawa E, Kimura M, Koike H, Ueno Y, *et al.* Vascularized and Complex Organ Buds from Diverse Tissues via Mesenchymal Cell-Driven Condensation. *Cell Stem Cell* 2015;16:556–565. doi: 10.1016/j.stem.2015.03.004.
 36. Kouji Y, Kido T, Ito T, Oyama H, Chen SW, Katou Y, *et al.* An In Vitro Human Liver Model by iPSC-Derived Parenchymal and Non-parenchymal Cells. *Stem Cell Reports* 2017;9:490–498. doi: 10.1016/j.stemcr.2017.06.010.
 37. Crosnier C, Artigues-Bitach T, Encha-Razavi F, Audollent S, Soudy F, Hadchouel M, *et al.* JAGGED1 gene expression during human embryogenesis elucidates the wide phenotypic spectrum of Alagille syndrome. *Hepatology* 2000;32:574–581. doi: 10.1053/jhep.2000.16600.
 38. Häberle J, Pauli S, Schmidt E, Schulze-Eilfing B, Berning C, Koch HG. Mild citrullinemia in Caucasians is an allelic variant of argininosuccinate synthetase deficiency (citrullinemia type 1). *Mol Genet Metab* 2003;80:302–306. doi: 10.1016/j.ymgme.2003.08.002.
 39. Akbari S, Sevinç GG, Ersoy N, Basak O, Kaplan K, Sevinç K, *et al.* Robust, Long-Term Culture of Endoderm-Derived Hepatic Organoids for Disease Modeling. *Stem Cell Reports* 2019;13:627–641. doi: 10.1016/j.stemcr.2019.08.007.
 40. Derks T, Rodriguez-Buritica DF, Ahmad A, de Boer F, Couce ML, Grünert SC, *et al.* Glycogen Storage Disease Type Ia: Current Management Options, Burden and Unmet Needs. *Nutrients* 2021;13:3828. doi: 10.3390/nu13113828.
 41. Mun SJ, Hong YH, Shin Y, Lee J, Cho HS, Kim DS, *et al.* Efficient and reproducible generation of human induced pluripotent stem cell-derived expandable liver organoids for disease modeling. *Sci Rep* 2023;13:22935. doi: 10.1038/s41598-023-50250-w.
 42. Kotronen A, Yki-Järvinen H. Fatty liver: A novel component of the metabolic syndrome. *Arterioscler Thromb Vasc Biol* 2008;28:27–38. doi: 10.1161/ATVBAHA.107.147538.
 43. Mun SJ, Ryu JS, Lee MO, Son YS, Oh SJ, Cho HS, *et al.* Generation of expandable human pluripotent stem cell-derived hepatocyte-like liver organoids. *J Hepatol* 2019;71:970–985. doi: 10.1016/j.jhep.2019.06.030.
 44. Ramli MNB, Lim YS, Koe CT, Demircioglu D, Tng W, Gonzales KAU, *et al.* Human Pluripotent Stem Cell-Derived Organoids as Models of Liver Disease. *Gastroenterology* 2020;159:1471–1486.e12. doi: 10.1053/j.gastro.2020.06.010.
 45. Feng Z, Zhou B, Shuai Q, Wei Y, Jin N, Wang X, *et al.* Development of an alcoholic liver disease model for drug evaluation from human induced pluripotent stem cell-derived liver organoids. *Acta Biochim Biophys Sin (Shanghai)* 2024;56:1460–1472. doi: 10.3724/abbs.2024074.
 46. Tsukuda S, Watashi K. Hepatitis B virus biology and life cycle. *Antiviral Res* 2020;182:104925. doi: 10.1016/j.antiviral.2020.104925.

47. Gural N, Mancio-Silva L, He J, Bhatia SN. Engineered Livers for Infectious Diseases. *Cell Mol Gastroenterol Hepatol* 2018;5:131–144. doi: 10.1016/j.jcmgh.2017.11.005.
48. Nie YZ, Zheng YW, Miyakawa K, Murata S, Zhang RR, Sekine K, *et al.* Recapitulation of hepatitis B virus-host interactions in liver organoids from human induced pluripotent stem cells. *EBioMedicine* 2018;35:114–123. doi: 10.1016/j.ebiom.2018.08.014.
49. Guo H, Liu D, Liu K, Hou Y, Li C, Li Q, *et al.* Drug repurposing screen identifies vidofludimus calcium and pyrazofurin as novel chemical entities for the development of hepatitis E interventions. *Virol Sin* 2024;39:123–133. doi: 10.1016/j.virs.2023.11.006.
50. Natarajan V, Simoneau CR, Erickson AL, Meyers NL, Baron JL, Cooper S, *et al.* Modelling T-cell immunity against hepatitis C virus with liver organoids in a microfluidic coculture system. *Open Biol* 2022;12:210320. doi: 10.1098/rsob.210320.
51. Chang M, Bogacheva MS, Lou YR. Challenges for the Applications of Human Pluripotent Stem Cell-Derived Liver Organoids. *Front Cell Dev Biol* 2021;9:748576. doi: 10.3389/fcell.2021.748576.
52. Sato T, Stange DE, Ferrante M, Vries R, Es J, Brink S, *et al.* Long-term Expansion of Epithelial Organoids From Human Colon, Adenoma, Adenocarcinoma, and Barrett's Epithelium. *Gastroenterology* 2011; 141:1762–1772. doi: 10.1053/j.gastro.
53. Lannagan T, Lee YK, Wang T, Roper J, Bettington ML, Fennell L, *et al.* Genetic editing of colonic organoids provides a molecularly distinct and orthotopic preclinical model of serrated carcinogenesis. *Gut* 2019;68:684–692. doi: 10.1136/gutjnl-2017-315920.
54. Drost J, van Jaarsveld RH, Ponsioen B, Zimmerlin C, van Boxtel R, Buijs A, *et al.* Sequential cancer mutations in cultured human intestinal stem cells. *Nature* 2015;521:43–47. doi: 10.1038/nature14415.
55. Nguyen R, Da Won Bae S, Qiao L, George J. Developing liver organoids from induced pluripotent stem cells (iPSCs): An alternative source of organoid generation for liver cancer research. *Cancer Lett* 2021;508:13–17. doi: 10.1016/j.canlet.2021.03.017.
56. Wang H, Calvisi DF, Chen X. Organoids for the Study of Liver Cancer. *Semin Liver Dis* 2021;41:19–27. doi: 10.1055/s-0040-1719176.
57. Artegiani B, van Voorthuijsen L, Lindeboom R, Seinstra D, Heo I, Tapia P, *et al.* Probing the Tumor Suppressor Function of BAP1 in CRISPR-Engineered Human Liver Organoids. *Cell Stem Cell* 2019;24:927–943.e6. doi: 10.1016/j.stem.2019.04.017.
58. Sun L, Wang Y, Cen J, Ma X, Cui L, Qiu Z, *et al.* Modelling liver cancer initiation with organoids derived from directly reprogrammed human hepatocytes. *Nat Cell Biol* 2019;21:1015–1026. doi: 10.1038/s41556-019-0359-5.
59. Guan Y, Chen X, Wu M, Zhu W, Arslan A, Takeda S, *et al.* The phosphatidylethanolamine biosynthesis pathway provides a new target for cancer chemotherapy. *J Hepatol* 2020;72:746–760. doi: 10.1016/j.jhep.2019.11.007.
60. Sampaziotis F, de Brito MC, Madrigal P, Bertero A, Saeb-Parsy K, Soares F, *et al.* Cholangiocytes derived from human induced pluripotent stem cells for disease modeling and drug validation. *Nat Biotechnol* 2015;33:845–852. doi: 10.1038/nbt.3275.
61. Ogawa M, Ogawa S, Bear CE, Ahmadi S, Chin S, Li B, *et al.* Directed differentiation of cholangiocytes from human pluripotent stem cells. *Nat Biotechnol* 2015;33:853–861. doi: 10.1038/nbt.3294.
62. Zabulica M, Jakobsson T, Ravaioli F, Vosough M, Gramignoli R, Ellis E, *et al.* Gene Editing Correction of a Urea Cycle Defect in Organoid Stem Cell Derived Hepatocyte-like Cells. *Int J Mol Sci* 2021;22:1217. doi: 10.3390/ijms22031217.
63. Ng SS, Saeb-Parsy K, Blackford S, Segal JM, Serra MP, Horcas-Lopez M, *et al.* Human iPSC derived progenitors bioengineered into liver organoids using an inverted colloidal crystal poly (ethylene glycol) scaffold. *Biomaterials* 2018;182:299–311. doi: 10.1016/j.biomaterials.2018.07.043.
64. Chen J, Shen L, Guo Q, Ma S, Zhang Y, Chen J, *et al.* The down-regulation of Tapasin in dendritic cell regulates CD8(+) T cell autophagy to hamper hepatitis B viral clearance in the induced pluripotent stem cell-derived hepatocyte organoid. *J Med Virol* 2024;96:e29546. doi: 10.1002/jmv.29546.
65. Yang L, Han Y, Nilsson-Payant BE, Gupta V, Wang P, Duan X, *et al.* A Human Pluripotent Stem Cell-based Platform to Study SARS-CoV-2 Tropism and Model Virus Infection in Human Cells and Organoids. *Cell Stem Cell* 2020;27:125–136.e7. doi: 10.1016/j.stem.2020.06.015.
66. Richards A, Friesen M, Khalil A, Barrasa MI, Gehrke L, Jaenisch R. SARS-CoV-2 infection of human pluripotent stem cell-derived liver organoids reveals potential mechanisms of liver pathology. *iScience* 2022;25:105146. doi: 10.1016/j.isci.2022.105146.
67. Li Y, Nie Y, Yang X, Liu Y, Deng X, Hayashi Y, *et al.* Integration of Kupffer cells into human iPSC-derived liver organoids for modeling liver dysfunction in sepsis. *Cell Rep* 2024;43:113918. doi: 10.1016/j.celrep.2024.113918.
68. Avior Y, Sagi I, Benvenisty N. Pluripotent stem cells in disease modelling and drug discovery. *Nat Rev Mol Cell Biol* 2016;17:170–182. doi: 10.1038/nrm.2015.27.
69. Caponigro G, Sellers WR. Advances in the preclinical testing of cancer therapeutic hypotheses. *Nat Rev Drug Discov* 2011;10:179–187. doi: 10.1038/nrd3385.
70. Heydari Z, Gramignoli R, Piryaee A, Zahmatkesh E, Pooyan P, Seydi H, *et al.* Standard Protocols for Characterising Primary and In Vitro-Generated Human Hepatocytes. *J Cell Mol Med* 2025;29:e70390. doi: 10.1111/jcmm.70390.
71. Zhang K, Zhang L, Liu W, Ma X, Cen J, Sun Z, *et al.* In Vitro Expansion of Primary Human Hepatocytes with Efficient Liver Repopulation Capacity. *Cell Stem Cell* 2018;23:806–819.e4. doi: 10.1016/j.stem.2018.10.018.
72. Poisson J, Lemoine S, Boulanger C, Durand F, Moreau R, Valla D, *et al.* Liver sinusoidal endothelial cells: Physiology and role in liver diseases. *J Hepatol* 2017;66:212–227. doi: 10.1016/j.jhep.2016.07.009.
73. Medine CN, Lucendo-Villarin B, Storck C, Wang F, Szkolnicka D, Khan F, *et al.* Developing high-fidelity hepatotoxicity models from pluripotent stem cells. *Stem Cells Transl Med* 2013;2:505–509. doi: 10.5966/sctm.2012-0138.
74. Ware BR, Berger DR, Khetani SR. Prediction of Drug-Induced Liver Injury in Micropatterned Co-cultures Containing iPSC-Derived Human Hepatocytes. *Toxicol Sci* 2015;145:252–262. doi: 10.1093/toxsci/kfv048.
75. Shinozawa T, Kimura M, Cai Y, Saiki N, Yoneyama Y, Ouchi R, *et al.* High-Fidelity Drug-Induced Liver Injury Screen Using Human Pluripotent Stem Cell-Derived Organoids. *Gastroenterology* 2021;160:831–846.e10. doi: 10.1053/j.gastro.2020.10.002.
76. Cheng W, Zhou Y, Xie Y, Li Y, Zhou R, Wang H, *et al.* Combined effect of polystyrene microplastics and bisphenol A on the human embryonic stem cells-derived liver organoids: The hepatotoxicity and lipid accumulation. *Sci Total Environ* 2023;854:158585. doi: 10.1016/j.scitotenv.2022.158585.
77. Kim H, Im I, Jeon JS, Kang EH, Lee HA, Jo S, *et al.* Development of human pluripotent stem cell-derived hepatic organoids as an alternative model for drug safety assessment. *Biomaterials* 2022;286:121575. doi: 10.1016/j.biomaterials.2022.121575.
78. Wu X, Jiang D, Yang Y, Li S, Ding Q. Modeling drug-induced liver injury and screening for anti-hepatofibrotic compounds using human PSC-derived organoids. *Cell Regen* 2023;12:6. doi: 10.1186/s13619-022-00148-1.
79. Harrison SP, Baumgarten SF, Chollet ME, Stavik B, Bhattacharya A, Almas R, *et al.* Parenteral nutrition emulsion inhibits CYP3A4 in an iPSC derived liver organoids testing platform. *J Pediatr Gastroenterol Nutr* 2024;78:1047–1058. doi: 10.1002/jpn.3.12195.
80. Sayed N, Liu C, Wu JC. Translation of Human-Induced Pluripotent Stem Cells: From Clinical Trial in a Dish to Precision Medicine. *J Am Coll Cardiol* 2016;67:2161–2176. doi: 10.1016/j.jacc.2016.01.083.
81. Shrestha S, Lekkala V, Acharya P, Kang SY, Vanga MG, Lee MY. Reproducible generation of human liver organoids (HLOs) on a pillar plate platform via microarray 3D bioprinting. *Lab Chip* 2024;24:2747–2761. doi: 10.1039/d4lc00149d.
82. Zhang CJ, Meyer SR, O'Meara MJ, Huang S, Capeling MM, Ferrer-Torres D, *et al.* A human liver organoid screening platform for DILI risk prediction. *J Hepatol* 2023;78:998–1006. doi: 10.1016/j.jhep.2023.01.019.
83. Guan Y, Enejder A, Wang M, Fang Z, Cui L, Chen SY, *et al.* A human multi-lineage hepatic organoid model for liver fibrosis. *Nat Commun* 2021;12:6138. doi: 10.1038/s41467-021-26410-9.
84. Sgodda M, Dai Z, Zweigerdt R, Sharma AD, Ott M, Cantz T. A Scalable Approach for the Generation of Human Pluripotent Stem Cell-Derived Hepatic Organoids with Sensitive Hepatotoxicity Features. *Stem Cells Dev* 2017;26:1490–1504. doi: 10.1089/scd.2017.0023.
85. Wang Y, Wang H, Deng P, Chen W, Guo Y, Tao T, *et al.* In situ differentiation and generation of functional liver organoids from human

- iPSCs in a 3D perfusable chip system. *Lab Chip* 2018;18:3606–3616. doi: 10.1039/c8lc00869h.
86. Lee H. Engineering In vitro Models: Bioprinting of Organoids with Artificial Intelligence. *Cyborg Bionic Syst* 2023;4:0018. doi: 10.34133/cbsystems.0018.
 87. Du X, Chen Z, Li Q, Yang S, Jiang L, Yang Y, *et al.* Organoids revealed: Morphological analysis of the profound next generation in-vitro model with artificial intelligence. *Biodes Manuf* 2023;6:319–339. doi: 10.1007/s42242-022-00226-y.
 88. Furuta T, Furuya K, Zheng YW, Oda T. Novel alternative transplantation therapy for orthotopic liver transplantation in liver failure: A systematic review. *World J Transplant* 2020;10:64–78. doi: 10.5500/wjt.v10.i3.64.
 89. Mito M, Kusano M, Kawaura Y. Hepatocyte transplantation in man. *Transplant Proc* 1992;24:3052–3053.
 90. Iansante V, Mitry RR, Filippi C, Fitzpatrick E, Dhawan A. Human hepatocyte transplantation for liver disease: Current status and future perspectives. *Pediatr Res* 2018;83:232–240. doi: 10.1038/pr.2017.284.
 91. Soltys KA, Setoyama K, Tafaleng EN, Soto Gutiérrez A, Fong J, Fukumitsu K, *et al.* Host conditioning and rejection monitoring in hepatocyte transplantation in humans. *J Hepatol* 2017;66:987–1000. doi: 10.1016/j.jhep.2016.12.017.
 92. Carpentier A, Tesfaye A, Chu V, Nimgaonkar I, Zhang F, Lee SB, *et al.* Engrafted human stem cell-derived hepatocytes establish an infectious HCV murine model. *J Clin Invest* 2014;124:4953–4964. doi: 10.1172/JCI75456.
 93. Blackford S, Ng SS, Segal JM, King A, Austin AL, Kent D, *et al.* Validation of Current Good Manufacturing Practice Compliant Human Pluripotent Stem Cell-Derived Hepatocytes for Cell-Based Therapy. *Stem Cells Transl Med* 2019;8:124–137. doi: 10.1002/scrm.18-0084.
 94. Nagamoto Y, Takayama K, Ohashi K, Okamoto R, Sakurai F, Tachibana M, *et al.* Transplantation of a human iPSC-derived hepatocyte sheet increases survival in mice with acute liver failure. *J Hepatol* 2016;64:1068–1075. doi: 10.1016/j.jhep.2016.01.004.
 95. Rashidi H, Luu NT, Alwahsh SM, Ginai M, Alhaque S, Dong H, *et al.* 3D human liver tissue from pluripotent stem cells displays stable phenotype in vitro and supports compromised liver function in vivo. *Arch Toxicol* 2018;92:3117–3129. doi: 10.1007/s00204-018-2280-2.
 96. Tsuchida T, Murata S, Matsuki K, Mori A, Matsuo M, Mikami S, *et al.* The Regenerative Effect of Portal Vein Injection of Liver Organoids by Retrorsine/Partial Hepatectomy in Rats. *Int J Mol Sci* 2019;21:178. doi: 10.3390/ijms21010178.
 97. Tsuchida T, Murata S, Hasegawa S, Mikami S, Enosawa S, Hsu HC, *et al.* Investigation of Clinical Safety of Human iPS Cell-Derived Liver Organoid Transplantation to Infantile Patients in Porcine Model. *Cell Transplant* 2020;29:963689720964384. doi: 10.1177/0963689720964384.
 98. Olgasi C, Cucci A, Follenzi A. iPSC-Derived Liver Organoids: A Journey from Drug Screening, to Disease Modeling, Arriving to Regenerative Medicine. *Int J Mol Sci* 2020;21:6215. doi: 10.3390/ijms21176215.
 99. Mun SJ, Lee J, Chung KS, Son MY, Son MJ. Effect of Microbial Short-Chain Fatty Acids on CYP3A4-Mediated Metabolic Activation of Human Pluripotent Stem Cell-Derived Liver Organoids. *Cells* 2021;10:126. doi: 10.3390/cells10010126.
 100. Zahmatkesh E, Ghanian MH, Zarkesh I, Farzaneh Z, Halvaei M, Heydari Z, *et al.* Tissue-Specific Microparticles Improve Organoid Microenvironment for Efficient Maturation of Pluripotent Stem-Cell-Derived Hepatocytes. *Cells* 2021;10:1274. doi: 10.3390/cells10061274.
 101. Liu K, Chen X, Fan Z, Ren F, Liu J, Hu B. From organoids to organoids-on-a-chip: Current applications and challenges in biomedical research. *Chin Med J* 2025;138:792–807. doi: 10.1097/CM9.0000000000003535.
 102. Utami T, Danoy M, Khadim RR, Tokito F, Arakawa H, Kato Y, *et al.* A highly efficient cell culture method using oxygen-permeable PDMS-based honeycomb microwells produces functional liver organoids from human induced pluripotent stem cell-derived carboxypeptidase M liver progenitor cells. *Biotechnol Bioeng* 2024;121:1178–1190. doi: 10.1002/bit.28640.
 103. Weng Y, Han S, Sekyi MT, Su T, Mattis AN, Chang TT. Self-Assembled Matrigel-Free iPSC-Derived Liver Organoids Demonstrate Wide-Ranging Highly Differentiated Liver Functions. *Stem Cells* 2023;41:126–139. doi: 10.1093/stmcls/sxac090.
 104. Byeon JH, Jung DJ, Han HJ, Son WC, Jeong GS. Fast formation and maturation enhancement of human liver organoids using a liver-organoid-on-a-chip. *Front Cell Dev Biol* 2024;12:1452485. doi: 10.3389/fcell.2024.1452485.
 105. Obeid DA, Mir TA, Alzhrani A, Altuhami A, Shamma T, Ahmed S, *et al.* Using Liver Organoids as Models to Study the Pathobiology of Rare Liver Diseases. *Biomedicines* 2024;12:446. doi: 10.3390/biomedicines12020446.
 106. Sharma S, Raju R, Sui S, Hu WS. Stem cell culture engineering - process scale up and beyond. *Biotechnol J* 2011;6:1317–1329. doi: 10.1002/biot.201000435.
 107. Horslen SP, Fox IJ. Hepatocyte transplantation. *Transplantation* 2004;77:1481–1486. doi: 10.1097/01.tp.0000113809.53415.c2.
 108. Xu X, Jiang S, Gu L, Li B, Xu F, Li C, *et al.* High-throughput bioengineering of homogenous and functional human-induced pluripotent stem cells-derived liver organoids via micropatterning technique. *Front Bioeng Biotechnol* 2022;10:937595. doi: 10.3389/fbioe.2022.937595.
 109. Salmon I, Grebenyuk S, Abdel Fattah AR, Rustandi G, Pilkington T, Verfaillie C, *et al.* Engineering neurovascular organoids with 3D printed microfluidic chips. *Lab Chip* 2022;22:1615–1629. doi: 10.1039/d1lc00535a.
 110. Nguyen V, Ye S, Gkouzioti V, van Wolferen ME, Yengej FY, Melkert D, *et al.* A human kidney and liver organoid-based multi-organ-on-a-chip model to study the therapeutic effects and biodistribution of mesenchymal stromal cell-derived extracellular vesicles. *J Extracell Vesicles* 2022;11:e12280. doi: 10.1002/jev2.12280.
 111. Telles-Silva KA, Pacheco L, Komatsu S, Chianca F, Caires-Júnior LC, Araujo B, *et al.* Applied Hepatic Bioengineering: Modeling the Human Liver Using Organoid and Liver-on-a-Chip Technologies. *Front Bioeng Biotechnol* 2022;10:845360. doi: 10.3389/fbioe.2022.845360.
 112. Meyer SR, Zhang CJ, Garcia MA, Procaro MC, Yoo S, Jolly AL, *et al.* A High-Throughput Microphysiological Liver Chip System to Model Drug-Induced Liver Injury Using Human Liver Organoids. *Gastro Hep Adv* 2024;3:1045–1053. doi: 10.1016/j.gastha.2024.08.004.
 113. Lekkala V, Shrestha S, Al Qaryoute A, Dhinoja S, Acharya P, Raheem A, *et al.* Enhanced Maturity and Functionality of Vascular Human Liver Organoids through 3D Bioprinting and Pillar Plate Culture. *ACS Biomater Sci Eng* 2025;11:506–517. doi: 10.1021/acsbomaterials.4c01658.
 114. Yamanaka S. Pluripotent Stem Cell-Based Cell Therapy-Promise and Challenges. *Cell Stem Cell* 2020;27:523–531. doi: 10.1016/j.stem.2020.09.014.
 115. Laurent LC, Ulitsky I, Slavin I, Tran H, Schork A, Morey R, *et al.* Dynamic changes in the copy number of pluripotency and cell proliferation genes in human ESCs and iPSCs during reprogramming and time in culture. *Cell Stem Cell* 2011;8:106–118. doi: 10.1016/j.stem.2010.12.003.
 116. Assou S, Girault N, Plinet M, Bouckenheimer J, Sansac C, Combe M, *et al.* Recurrent Genetic Abnormalities in Human Pluripotent Stem Cells: Definition and Routine Detection in Culture Supernatant by Targeted Droplet Digital PCR. *Stem Cell Reports* 2020;14:1–8. doi: 10.1016/j.stemcr.2019.12.004.
 117. Sekine K, Ogawa S, Tsuzuki S, Kobayashi T, Ikeda K, Nakanishi N, *et al.* Generation of human induced pluripotent stem cell-derived liver buds with chemically defined and animal origin-free media. *Sci Rep* 2020;10:17937. doi: 10.1038/s41598-020-73908-1.
 118. Klotz BJ, Oosterhoff LA, Utomo L, Lim KS, Vallmajo-Martin Q, Clevers H, *et al.* A Versatile Biosynthetic Hydrogel Platform for Engineering of Tissue Analogues. *Adv Healthc Mater* 2019;8:e1900979. doi: 10.1002/adhm.201900979.
 119. Ye S, Boeter J, Mihajlovic M, van Steenbeek FG, van Wolferen ME, Oosterhoff LA, *et al.* A Chemically Defined Hydrogel for Human Liver Organoid Culture. *Adv Funct Mater* 2020;30:2000893. doi: 10.1002/adfm.202000893.
 120. Krüger M, Oosterhoff LA, van Wolferen ME, Schiele SA, Walther A, Geijsen N, *et al.* Cellulose Nanofibril Hydrogel Promotes Hepatic Differentiation of Human Liver Organoids. *Adv Healthc Mater* 2020;9:e1901658. doi: 10.1002/adhm.201901658.
 121. Roudaut M, Caillaud A, Souguir Z, Bray L, Girardeau A, Rimbart A, *et al.* Human induced pluripotent stem cells-derived liver organoids grown on a Biomimesys® hyaluronic acid-based hydroscaffold as a new model for studying human lipoprotein metabolism. *Bioeng Transl Med* 2024;9:e10659. doi: 10.1002/btm2.10659.

122. Wang Y, Liu H, Zhang M, Wang H, Chen W, Qin J. One-step synthesis of composite hydrogel capsules to support liver organoid generation from hiPSCs. *Biomater Sci* 2020;8:5476–5488. doi: 10.1039/d0bm01085e.
123. Zhu X, Zhang B, He Y, Bao J. Liver Organoids: Formation Strategies and Biomedical Applications. *Tissue Eng Regen Med* 2021;18:573–585. doi: 10.1007/s13770-021-00357-w.
124. Xu H, Wang B, Ono M, Kagita A, Fujii K, Sasakawa N, *et al.* Targeted Disruption of HLA Genes via CRISPR-Cas9 Generates iPSCs with Enhanced Immune Compatibility. *Cell Stem Cell* 2019;24:566–578.e7. doi: 10.1016/j.stem.2019.02.005.
125. Koga K, Wang B, Kaneko S. Current status and future perspectives of HLA-edited induced pluripotent stem cells. *Inflamm Regen* 2020;40:23. doi: 10.1186/s41232-020-00132-9.
126. Bi G, Zhang X, Li W, Lu X, He X, Li Y, *et al.* Modeling alcohol-associated liver disease in humans using adipose stromal or stem cell-derived organoids. *Cell Rep Methods* 2024;4:100778. doi: 10.1016/j.crmeth.2024.100778.
127. Huang P, Zhang L, Gao Y, He Z, Yao D, Wu Z, *et al.* Direct reprogramming of human fibroblasts to functional and expandable hepatocytes. *Cell Stem Cell* 2014;14:370–384. doi: 10.1016/j.stem.2014.01.003.
128. Kim J, Koo BK, Knoblich JA. Human organoids: model systems for human biology and medicine. *Nat Rev Mol Cell Biol* 2020;21:571–584. doi: 10.1038/s41580-020-0259-3.
129. Sang C, Lin J, Ji S, Gao Q. Progress, application and challenges of liver organoids. *Clinical Cancer Bulletin* 2024;3:1–17. doi: 10.1007/s44272-024-00012-0.

How to cite this article: Feng ZW, Zhou BR, Shuai QZ, Mu JB, Xie J. Applications and limitations of pluripotent stem cell-derived liver organoids. *Chin Med J* 2026;139:362–375. doi: 10.1097/CM9.00000000000003707